

Score: / 10

Calculus I (MATH 2200)
Spring 2020
Wednesday, 02/05/2020
Quiz 2: Section 2.3, 2.4

Name (Print): _____

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Section 2.3

1. Given that $\lim_{x \rightarrow 2} f(x) = 1$ and $\lim_{x \rightarrow 2} g(x) = -11$, find the indicated limits.

2 pts

(a) $\lim_{x \rightarrow 2} 2g(x) + 5f(x)$

$$\begin{aligned}\lim_{x \rightarrow 2} 2g(x) + 5f(x) &= \lim_{x \rightarrow 2} 2g(x) + \lim_{x \rightarrow 2} 5f(x) \\ &= 2 \lim_{x \rightarrow 2} g(x) + 5 \lim_{x \rightarrow 2} f(x) \\ &= 2(-11) + 5(1) = -17\end{aligned}$$

2 pts

(b) $\lim_{x \rightarrow 2} \frac{2x - f(x)}{x^2 + 11}$

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{2x - f(x)}{x^2 + 11} &= \frac{\lim_{x \rightarrow 2} (2x - f(x))}{\lim_{x \rightarrow 2} (x^2 + 11)} \\ &= \frac{\lim_{x \rightarrow 2} 2x - \lim_{x \rightarrow 2} f(x)}{\lim_{x \rightarrow 2} x^2 + \lim_{x \rightarrow 2} 11} \\ &= \frac{4 - 1}{4 + 11} \\ &= \frac{3}{15} \\ &= \frac{1}{5}\end{aligned}$$

2 pts

2. Suppose f satisfies $2 \leq f(x) \leq 1 + x^2$ for all values of x near 1. Find $\lim_{x \rightarrow 1} f(x)$.

If f satisfies $2 \leq f(x) \leq 1 + x^2$ for all values of x near 1, then notice that

$$\lim_{x \rightarrow 1} 2 = 2 \text{ and } \lim_{x \rightarrow 1} 1 + x^2 = 1 + 1^2 = 2.$$

As such, the Squeeze Theorem then yields that that

$$\lim_{x \rightarrow 1} f(x) = 2.$$

2 pts

3. Let $f(x) = \frac{|x - 2|}{x - 2}$. Explain why the limit $\lim_{x \rightarrow 2} f(x)$ does not exist.

First, we calculate the left-hand and right-hand limits:

1. As x approaches 2 from the right side, the value of the numerator will be positive and the value of the denominator will be positive (because the x -values are larger than 2). So, $\lim_{x \rightarrow 2^+} f(x) = 1$.
2. As x approaches 2 from the left side, the value of the numerator will be positive and the value of the denominator will be negative (because the x -values are smaller than 2). So, $\lim_{x \rightarrow 2^-} f(x) = -1$.

Hence, as the one-sided limits are not equal to the same value, the two-sided limit does not exist.

Section 2.4

2 pts

1. Analyze
- $\lim_{x \rightarrow -4^+} \frac{-x^3 + 5x^2 - 6x}{-x^3 - 4x^2}$
- .

To analyze $\lim_{x \rightarrow -4^+} \frac{-x^3 + 5x^2 - 6x}{-x^3 - 4x^2}$, we analyze the numerator and denominator of the function in question. But first, let's make our lives easier by simplifying. Observe that

$$\begin{aligned} \frac{-x^3 + 5x^2 - 6x}{-x^3 - 4x^2} &= \frac{-x(x-2)(x-3)}{-x^2(x+4)} \\ &= \frac{(x-2)(x-3)}{x(x+4)}. \end{aligned}$$

So, let us now note that the numerator, $(x-2)(x-3)$, is positive and approaches 42 as x approaches -4 from the right. Then, note that the denominator, $x(x+4)$, is negative and approaches 0 as x approaches -4 from the right. So, $\lim_{x \rightarrow -4^+} \frac{-x^3 + 5x^2 - 6x}{-x^3 - 4x^2} = -\infty$.

2 (bonus)

2. Analyze
- $\lim_{x \rightarrow 0^+} \frac{x+5}{x}$
- .

To analyze $\lim_{x \rightarrow 0^+} \frac{x+5}{x}$, we analyze the numerator and denominator of the function in question. First, note that the numerator, $(x+5)$, is positive and approaches 5 as x approaches 0 from the right. Then, note that the denominator, x , is positive and approaches 0 as x approaches 0 from the right. So, $\lim_{x \rightarrow 0^+} \frac{x+5}{x} = \infty$.

1 (bonus)

3. Analyze
- $\lim_{x \rightarrow 0^+} \frac{1}{x}$
- .

To analyze $\lim_{x \rightarrow 0^+} \frac{1}{x}$, we analyze the numerator and denominator of the function in question. First, note that the numerator is constant and positive. Then, note that the denominator, x , is positive and approaches 0 as x approaches 0 from the right. So, $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$.